



f-RecX: A framework for designing effective textual explanations in recommender systems' user interfaces[☆]

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ABSTRACT

Recommender systems (RecSys) have become ubiquitous in users' daily digital interactions, significantly influencing decision-making processes. As these systems grow in algorithmic complexity, effective explanations for non-expert users become essential to fostering understanding and trust. While academic research explores diverse explanation methods, commercial applications predominantly employ textual explanations due to their implementation efficiency and user familiarity. However, the effectiveness of these textual explanations is often compromised by suboptimal presentation within RecSys user interfaces (UIs), leading to reduced user engagement and comprehension. This issue is particularly relevant given the recent emergence of large language models (LLMs) for generating RecSys explanations. We introduce f-RecX, a conceptual framework for characterizing and designing effective textual explanations in RecSys UIs. Based on a two-phase methodology combining qualitative user studies and quantitative evaluations, f-RecX maps four input dimensions (Explanation Style, Goals, Domain Dynamics, and Recommender Systems Technique) to an output dimension focused on visual representation. The framework aims to enhance the 'consumability' of textual explanations by making them easier to locate and comprehend, and more valuable for non-expert users. We demonstrate f-RecX's applicability through a usage scenario and analysis of existing RecSys UIs, offering valuable insights for enhancing explainability and user experience.

1. Introduction

Recommender systems (RecSys) have become an integral part of users' digital experiences, guiding users' choices in e-commerce, entertainment, social media, and beyond. These systems use algorithms to suggest items to users based on their preferences and behavior (Ricci et al., 2021; Jannach et al., 2010). As these systems grow more complex, the need for transparency and user understanding of model decisions has become more relevant than ever. Explanations in RecSys can improve trust, satisfaction, and user experience (UX) by connecting complex algorithms with user comprehension (Zhang et al., 2020; Tintarev and Masthoff, 2015).

While various forms of explanations exist, academic and commercial approaches have diverged. Academic research explores a wide range of methods, including sophisticated visual representations and interactive interfaces (Cardoso et al., 2019; Chatti et al., 2024; He et al., 2016), allowing users to explore different aspects of recommendations. These

methods often aim to provide deeper insights into the recommendation process and empower users with greater control. In contrast, commercial applications predominantly use textual explanations due to their ease of implementation, scalability, and user familiarity with text-based information.

Despite the prevalence of textual explanations in commercial applications, general principles for designing and presenting them in RecSys user interfaces (UIs) remain largely unexplored. This gap is evidenced by several indicators from recent literature. First, comprehensive reviews of explainable RecSys by Zhang et al. (2020) and Ko et al. (2022) focus predominantly on algorithmic explanation generation methods, dedicating minimal attention to UI presentation factors. Second, major commercial platforms like Netflix, Spotify, and Amazon employ widely varying textual explanation designs without established best practices. Third, existing XAI frameworks (Jin et al., 2021; Herm et al., 2022) provide general design principles but lack specific guidance for textual

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explanation visual characteristics, including font sizing, positioning, and contrast requirements. This creates a disconnect where RecSys researchers understand what to explain but lack guidance on how to present explanations effectively, while UI researchers provide general principles but lack domain-specific validation for recommendation contexts.

Furthermore, the increasing adoption of Large Language Models (LLMs) in generating recommendations and explanations (Castelnovo et al., 2024; Silva et al., 2024; Sun et al., 2024) underscores the importance of textual content in explanations. While LLM-generated explanations show promise (Lubos et al., 2024; Shi et al., 2024), we argue that their effectiveness ultimately depends on the visual presentation within the UI, regardless of their content quality. To address these challenges, we present f-RecX, a comprehensive theoretical framework for designing effective textual explanations in RecSys UIs.

While many frameworks for explainable AI (XAI) exist in the literature, they typically focus either on technical aspects of recommendations (Jin et al., 2021; Eiband et al., 2018) or human-centered design principles for XAI (Liao et al., 2020; Wang et al., 2019) in isolation. This fragmentation makes it challenging for designers and developers to gain a comprehensive overview of all relevant considerations when creating explainable RecSys applications. What is notably absent is an integrated framework that bridges technical recommendation techniques with human factors and visual design principles, specifically for textual explanations in recommender UIs.

To address these challenges, we present f-RecX, a comprehensive theoretical framework that uniquely combines RecSys' technical aspects and human-centered design considerations into a single integrated approach for designing effective textual explanations. We focus on textual explanations, aligning with dominant commercial trends. These explanations are well-established in RecSys UIs and offer a foundation for enhancement. By examining how to optimize this familiar format, we aim to improve user understanding and engagement. Moreover, we emphasize the importance of incorporating human-in-the-loop approaches throughout the design process, ensuring that explanations are not only algorithmically sound but also align with users' cognitive processes and information needs.

The identified gap between explanation content generation and effective presentation leads naturally to three interconnected research questions (RQ) that build upon each other:

RQ1 | How do non-expert users interact with and perceive recommender systems?

RQ2 | What visual characteristics of textual explanations most significantly impact user understanding in explainable recommender system interfaces?

RQ3 | How can the visual characteristics of textual explanations be optimized for explainable recommender system user interfaces?

RQ1 will help us to understand how users engage with textual explanations in their daily interactions with RecSys. This addresses the user-centric aspect of the research problem by establishing baseline user behavior and preferences. RQ2 addresses the core of the research gap by systematically investigating the visual characteristics that academic literature has not yet examined but that commercial implementations vary widely in their approach to. RQ3 looks at the practical solution to the identified gap by creating actionable design guidance that bridges academic insight with commercial implementation needs.

To answer these RQs, we combined qualitative user studies and quantitative evaluations to develop a structured methodology for tailoring explanations to RecSys's specific domain, user needs, and interface constraints, thereby bridging the gap between theoretical insights and practical implementation. Our primary contribution is f-RecX, shown in Fig. 1. f-RecX is a conceptual framework that encourages reflection and fosters the development of approaches for explainable RecSys. Through

the development process of f-RecX, we provide designers and developers with actionable guidelines for creating more human-centered, explainable RecSys and UIs.

This paper is structured as follows: Section 2 reviews related work in explainable RecSys and UI design for explanations. Section 3 details our methodology. Section 4 presents the analysis and results of the data collected. Section 5 presents our proposed framework, f-RecX. Section 6.2 demonstrates the applicability of f-RecX. Finally, Section 7 discusses the limitations and implications of our findings and outlines directions for future research.

2. Related work

In this section, we examine previous work on frameworks and guidelines for XAI UIs and previous studies on RecSys' explainability aspect. For a complete overview of the field of explainable RecSys, we recommend the work of Ko et al. (2022), Zhang et al. (2020), Tintarev and Masthoff (2015), Chatti et al. (2024), and Pu and Chen (2006).

2.1. Frameworks and guidelines for explainable AI user interfaces

Explainable Artificial Intelligence (XAI) emerged as a critical field addressing the opacity of advanced AI systems (Das and Rad, 2020; Adadi and Berrada, 2018; Gunning et al., 2019; Confalonieri et al., 2021). While significant progress has been made in developing algorithmic techniques for generating explanations, the user interface (UI) through which explanations are communicated remains a key issue. The UI serves as the mediator between the complex algorithm explanation and human understanding. Chromik and Butz (2021) note that the call for explanation should not be limited to providing post-hoc interpretation for an AI system's output. The UI must present explanations that enable users to understand, scrutinize, and potentially contest the AI's decisions. XAI UIs must be tailored to the needs of specific user groups in mind. Suresh et al. (2021) emphasize the importance of considering users' knowledge and goals when designing explanations for AI systems. Liao et al. (2020) propose a question-driven approach to guide UI designers on possible questions users are likely to ask when interacting with AI systems. Springer and Whittaker (2019) recommend progressive disclosure in XAI interfaces to prevent cognitive overload. In recent years, research on XAI UIs has evolved from foundational concepts to structured frameworks. Eiband et al. (2018) proposed a stage-based participatory design approach, while later works like Jin et al.'s EUCA framework (Jin et al., 2021) and Herm et al. (2022) theory offers a more concrete implementation guidance. Despite this progress, considerable limitations remain in XAI UI research. Most frameworks offer little concrete guidance on translating technical outputs into intuitive UI elements, leaving designers without clear direction on visual presentation parameters, especially when it comes to textual features inside a UI. Limited research exists on how to design XAI UIs for specific domains and user groups with different levels of AI literacy. Most research focuses on visual explanations while neglecting the design of textual explanations. Textual explanations are often more practical for implementation and, with the rise of LLMs, are becoming increasingly important for real-world applications. These gaps underscore the need for an integrated approach to XAI UI design that considers not only algorithmic characteristics but also human-centered aspects, providing guidance on the visual design of explanations.

2.2. Studies on explainable recommender systems

Explainability in RecSys has gained significant research interest, driven by the need for transparency, fairness, and user trust (Chatti et al., 2024; He et al., 2016; Zhang et al., 2020; Chen et al., 2022; Vultureanu-Albiși and Bădică, 2021; Gedikli et al., 2014). Although considerable progress has been made on individual aspects, the literature remains fragmented, and a critical yet often overlooked factor in

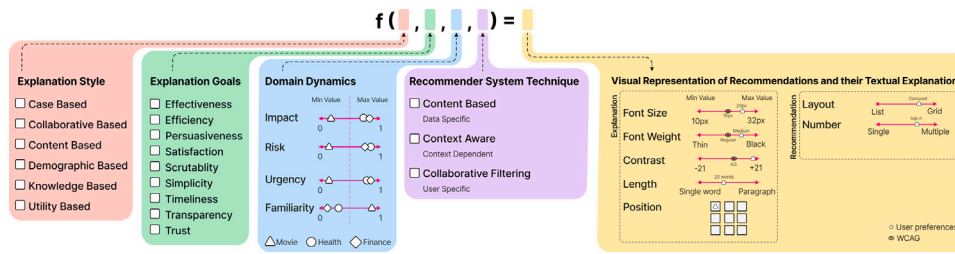


Fig. 1. The **f-RecX** framework consists of four input dimensions: Explanation Style, Explanation Goals, Domain Dynamics, and Recommender Systems Technique, and one output dimension: Visual Representation of Recommendations and their Textual Explanation. Each dimension is characterized by either a spectrum or a set of categories, reflecting different aspects of the field of explainable recommender systems. The output dimension displays the user preferences for a movie RecSys and general text accessibility guidelines. The framework is described in detail in Section 5.

this domain is the role of the UI (Beel and Dixon, 2021; Murphy-Hill and Murphy, 2013).

Early research focused predominantly on isolated explanation types; for instance, Herlocker et al. (2004) and Vig et al. (2009) investigated user-based and tag-based explanations, respectively. However, these contributions tend to address only one dimension of explainability, leaving a gap for comprehensive frameworks that integrate multiple aspects of UI design, algorithmic transparency, and user interaction.

More recent research has aimed at integrating multiple explanation modalities to overcome the limitations of single-aspect approaches (Kouki et al., 2020; Bhattacharya et al., 2023). For example, Kouki et al. (2020) introduced a hybrid recommender framework that generates personalized explanations by combining textual and visual methods. Their empirical evaluation suggests that no single explanation style universally outperforms the others; instead, effectiveness may depend on the application context and the user's needs. Similar integrated efforts are observed in model-agnostic approaches, such as XRec, which leverages large language models to produce more comprehensive, user-centered explanations (Ma et al., 2024).

Interactive approaches also play a critical role in evolving explainable RecSys (Al-Hazwani et al., 2021; Guesmi et al., 2024; Ooge et al., 2022; Szymanski et al., 2024). Guesmi et al. (2024) explored an interface that provides interactive explanations with multiple levels of detail in the context of scientific literature recommendations. Their work highlights the importance of giving users control over how much explanatory information they view, which can enhance transparency, trust, and overall UX.

These works suggest the need for a more systematic framework that consolidates the strengths of prior approaches while addressing common limitations, such as the lack of studies on the impact of UI elements and insufficient user involvement.

In summary, while existing research has significantly advanced our understanding of individual explanation strategies, a need remains for comprehensive frameworks. Such frameworks are critical for ensuring that explainable RecSys fulfills both technical and human-centric goals, thereby enabling better system transparency, trust, and overall user satisfaction.

3. Methodology

To address our research questions (see Section 1), we employed a two-step research methodology, with each step building upon the insights and findings of the previous one. This approach allowed us to systematically explore the design and perception of textual explanations in RecSys UIs. Fig. 2 provides an overview of the two-step methodology and its alignment with the research questions. The first method (M01) examined how non-expert users perceive RecSys UIs containing textual explanations through an empathy workshop. This qualitative and quantitative evaluation focused on understanding user preferences and identifying key elements needed to design effective textual explanations in a UI (RQ1 and RQ2). Building directly on these

findings, the second method (M02) involved an online survey designed to explore how specific visual characteristics of textual explanations impact UX and engagement (RQ3). This assessment evaluated the visual aspects identified as important during the empathy workshop.

The methodology is described in detail in the following subsections, including step-specific objectives. The findings are presented in Section 4.

3.1. M01 — Empathy workshop: Quantitative and qualitative assessment of commercial recommender systems user interfaces

The first method aimed to understand how non-expert users interact with and “consume” textual explanation within commercial RecSys UIs. In particular, we examined the multifaceted nature of these interactions through different lenses: identification of usage patterns, risk-impact assessment, quantitative evaluation of existing interfaces, and qualitative exploration of design preferences. These complementary perspectives provide a comprehensive understanding of the complex relationship between users and explanation interfaces, necessitating our four-objective approach, described below. We deliberately decided to focus on non-expert users, who are defined as individuals who may or may not have a technical background (Hohman et al., 2018), as they represent the primary audience for RecSys in real-world applications. Our study centered on commercial UIs due to participants’ potential familiarity, the predominant use of textual explanations, and public availability. It also represented systems that millions of users interact with daily, thereby maximizing the ecological validity and practical relevance of our findings.

We aimed to study four objectives (Obj.):

Obj. 1 | Identify commonly used RecSys-driven applications among non-expert users.

Obj. 2 | Assess users’ perceived impact-risk associated with accepting recommendations.

Obj. 3 | Collect quantitative, user-centered assessments of explanations in selected RecSys UIs.

Obj. 4 | Understand user preferences regarding UI design aspects of RecSys explanations.

These objectives are set in service of answering RQ1 and RQ2 (Section 1). To achieve these, we conducted a two-hour empathy workshop (Kaplan, 2019; Drouet et al., 2023; Ghajargar et al., 2017) with 19 participants. The workshop consisted of four activities, each addressing a distinct objective, with both individual and group tasks. The authors were available to answer participants’ questions, observe them, and take notes on insights gained during the group tasks.

3.1.1. Participant recruitment

Participants were recruited through the university network and announcements. The university network refers to the authors’ academic and social connections, including colleagues, students, and collaborators. In addition to personal outreach, announcements were shared via

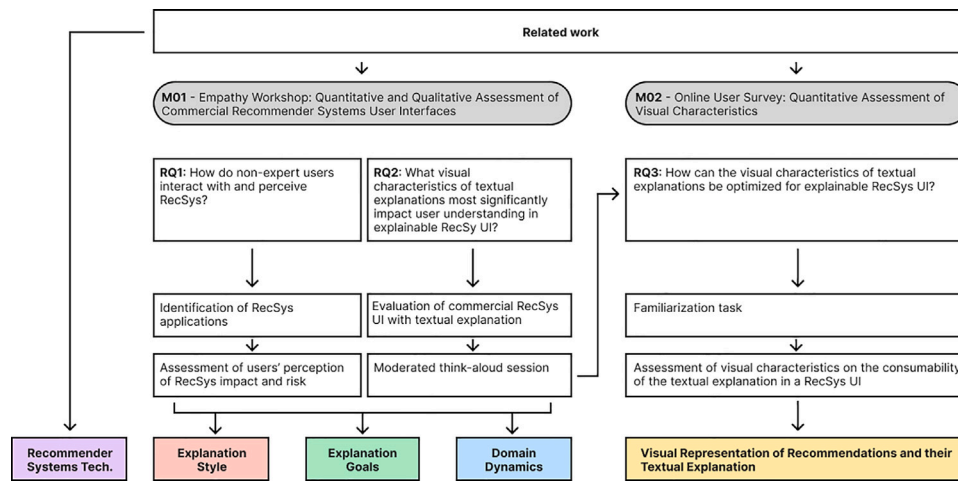


Fig. 2. This methodology overview aligns the two main methods (M) of the study with our three research questions (RQ) and their impact on the development of f-RecX dimensions. M01 involves an empathy workshop that assesses user perceptions and conducts user-centered evaluations of explanations. M02 involves a within-subject online survey that assesses the impact of visual characteristics on user experience and engagement in textual explanations.

university mailing lists, student platforms, and internal Slack channels. These announcements functioned similarly to recruitment ads, including a brief description of the study, its estimated duration, the types of activities involved, and assurances about data privacy and voluntary participation. No prior knowledge of RecSys, XAI, or UI design was required to avoid expertise bias, ensuring that participants would engage with the study from a layperson's perspective. This decision aligns with our interest in understanding how average, non-expert users interact with and interpret system outputs. As such, the inclusion criteria focused on people who self-identified as non-experts in the relevant fields. Participants were either curious about the topic, interested in contributing to research, or had spare time and volunteered after seeing the announcement on campus. This casual and low-barrier recruitment strategy allowed us to reach a broad and diverse set of participants. In line with ethical research practices and data minimization principles (Biega et al., 2020), we did not collect detailed demographic information such as age, education level, or occupation. However, based on recruitment channels, we assume that the majority of participants' origin is within Europe. While we did not formally collect gender data, a rough estimate based on self-identification during the sessions suggests a relatively balanced gender distribution. This estimation is provided to give some contextual understanding of the participant pool without violating privacy or collecting unnecessary personal data.

3.1.2. Workshop activities

Below is a description of the various activities and their relationship with the objectives, following the workshop's consecutive order.

Activity 1: Identification of Recommender System Applications (Obj. 1) | To identify commonly used RecSys-driven applications, participants engaged in a *self-reflection* activity. After introducing RecSys concepts and providing examples, participants individually noted on sticky notes: (i) The application name of RecSys they interact with; (ii) The recommendation data type (e.g., text, image, video, or audio); and (iii) The level of detail in the recommendation (e.g., only item, item + attribute features, or only item feature).

Activity 2: Risk-Impact Assessment (Obj. 2) | To assess users' perception of risks and impacts of accepting recommendations, we conducted a group activity using an impact-risk coordinate system. The coordinate system represented risk (potential negative outcomes) and impact (degree of influence on users). The choice of these two axes was based on XAI research (Molnar, 2020) and RecSys user perception studies (Pu and Chen, 2006). Then, in groups of 4–5, participants positioned their sticky notes from activity 1 on an A1-size coordinate system

with impact-risk axes, both ranging between -1 (low) and 1 (high). To foster discussion and discourse, we asked the group participants to position their sticky notes one after another and justify their positioning decisions within their group. Of particular interest were the RecSys-driven applications gathered by several participants, which facilitated the resolution of positioning conflicts.

Activity 3: UI Evaluation (Obj. 3) | To collect user-centered assessments of commercial explainable RecSys UIs, participants evaluated five UIs shown in Fig. 3 using a 7-point Likert scale for *usefulness*, *satisfaction*, and *ease of finding*. Usefulness and satisfaction are recognized as measurements to evaluate explanations (Colin et al., 2022; Gedikli et al., 2014; Ribeiro et al., 2016; Tintarev and Masthoff, 2015) and more generally RecSys (Silveira et al., 2019; Nanou et al., 2010; Knijnenburg et al., 2012). Usefulness measures how well the explanation aids understanding (Colin et al., 2022; Ribeiro et al., 2016), satisfaction assesses alignment with user expectations (Tintarev and Masthoff, 2007; Gedikli et al., 2014). We introduced ease of finding to evaluate the effectiveness and clarity of locating explanations in the UI.

Activity 4: Think-Aloud Session (Obj. 4) | To understand how current RecSys explanations reflect user preferences and needs, we conducted a moderated think-aloud session (Van Someren et al., 1994). Participants shared their assessments of the five UIs and discussed their need for explanation. Guiding questions included: “What makes an explanation useful?”, “How important are explanations in a UI?”, and “Should explanations be personalized, and would this raise privacy concerns?”

The insights gathered from these workshop activities directly informed the design of our online survey (M02), ensuring that our quantitative assessment focused on the visual characteristics that users identified as most relevant to the effectiveness of explanations. This phase also highlighted the importance of a human-in-the-loop approach in the early phase of designing explainable RecSys UIs. Moreover, this method motivates the inclusion of a human-centered dimension (*Domain Dynamics*) in the f-RecX framework (Section 5).

3.2. M02 — Online user survey: Quantitative assessment of visual characteristics

Building directly on insights about explanation visibility from the empathy workshop, we conducted an online survey to systematically explore how six visual characteristics influence the UX of textual explanations in RecSys UIs. The workshop findings helped us identify which visual elements were most important to users and informed

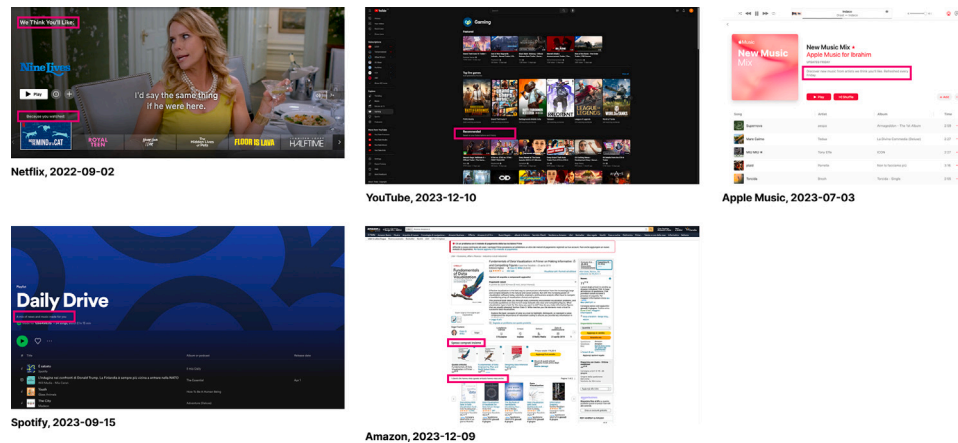


Fig. 3. Provided commercial RecSys UIs that contain an explanation with the retrieval date of the UI. These UIs are from various domains, including streaming services like Netflix and Spotify, as well as e-commerce platforms like Amazon. The pink stroke indicates the explanations provided by these RecSys-driven applications to non-expert users. These explanations are text-based, vary in location within the UI, and have different visual characteristics.

our selection of variables to test, including presentation factors and placement preferences. Building upon this, the M02 online survey aimed to optimize the presentation of textual explanations to enhance ‘consumability’ (RQ3). With the term consumability, we refer to the ease with which users can locate, understand, engage with, and derive value from the explanation.

3.2.1. Participants and recruitment

We recruited 61 participants through the authors’ personal and professional networks, as well as social media postings. This enabled us to engage with a diverse group of participants from both academic and non-academic backgrounds. The social media channels primarily included Slack, Twitter, and LinkedIn. For example, a typical Slack post included a brief description of the study, the estimated time commitment (e.g., “20 min”), a link to the survey or study platform, and reassurances about anonymity and voluntary participation.

No specific technical background was required for participation, which allowed us to include a broad range of perspectives and maintain accessibility for non-experts. Inclusion was based solely on interest and availability, not on professional experience or domain knowledge.

In line with ethical research practices and data minimization principles (Biega et al., 2020), we did not collect detailed demographic information. However, we estimate that most participants were based in Europe and had a mix of nationalities and backgrounds.

Participation was entirely voluntary, and no monetary or material compensation was provided.

Additionally, we were mindful of concerns about data quality in crowdsourcing raised by Iren and Bilgen (2014) and therefore prioritized quality over quantity. To ensure data quality, we implemented the following measures. We included an attention-checking question at the beginning of the survey and manually reviewed incomplete or inconsistent responses, which resulted in the exclusion of 5 entries that did not meet our minimum criteria. This approach helped us collect thoughtful responses while avoiding the noise and low-effort submissions sometimes found in open crowdsourced datasets.

3.2.2. Survey design and implementation

The survey consisted of 32 questions across seven sections, each designed to assess how changes in a single visual characteristic affected the perceived effectiveness of textual explanations. The survey was implemented using Qualtrics and took approximately 15–20 min to complete. The survey was pilot-tested with three users before deployment to identify any usability issues or unclear instructions. The final version of the survey incorporated refinements based on their feedback.

The overall completion rate was 62%, with 38 participants completing the entire survey.

Visual Characteristics Studied in the Survey | Based on the empathy workshop findings and established visual design principles, we selected six visual characteristics to evaluate textual explanations:

- Font size — ranging from 10 px to 32 px, with 16 px as the reference point (following WCAG guidelines¹)
- Font weight — from thin to black (9 weights of the Inter font family)
- Contrast — various background-text color combinations with different contrast ratios
- Position — six different placements of the explanation within the UI
- Number of recommendations — single item, top-n items, or m items (where $n < m$)
- Length of explanation — from a single word to a full paragraph

We consistently used the Inter font family across all conditions, as it is an accessible, open-source font designed specifically for computer screens and widely used in commercial applications.²

Survey Structure | The survey started with an informed consent form, followed by a brief introduction to the study’s goals and structure. The seven sections were structured as follows. In section one, *Familiarization Task*, participants were introduced to a RecSys UI resembling popular movie RecSys like Netflix or Amazon Prime (Fig. 4). Participants were asked to locate and transcribe the textual explanation within the UI to ensure they understood what constituted an explanation. In the next section, *Font Size Assessment*, participants viewed seven UIs with explanation font sizes ranging from 10 px to 32 px. In the third section, *Font Weight Assessment*, participants evaluated seven UIs with varying font weights from thin to black. In the fourth section, *Contrast Assessment*, six UIs with different background-text color combinations were presented, ranging from white background with black text to black background with white text. In the fifth section, *Position Assessment*, six UIs with explanations in different locations were shown, following a clockwise rotation pattern around the screen. In the second-to-last section, *Number of Recommendations Assessment*, three UIs displaying different numbers of recommendations alongside explanations were presented. In the last section, *Explanation Length Assessment*, six UIs with explanations of varying lengths, from a single word to a paragraph, were evaluated.

¹ <https://www.w3.org/WAI/standards-guidelines/wcag/>.

² <https://fonts.google.com/specimen/Inter/about>.

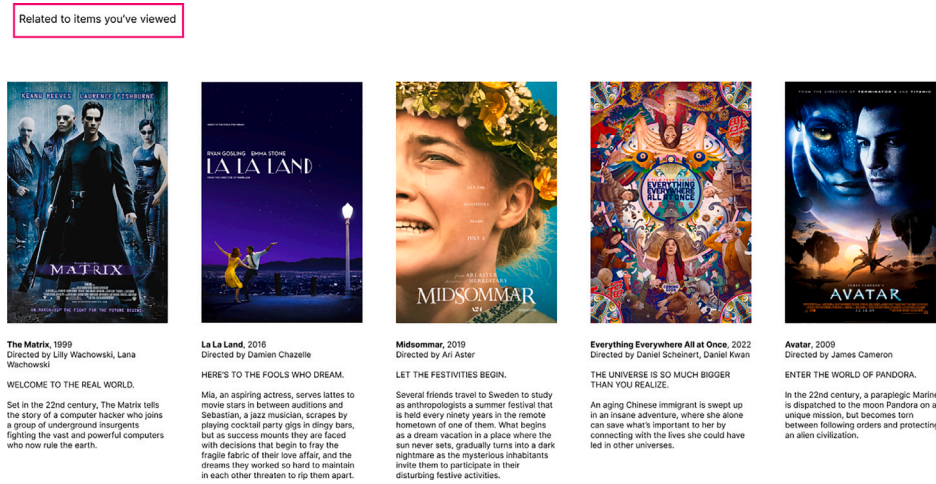


Fig. 4. UI designed for the familiarization task. The UI resembles a typical movie recommendation system interface, similar to those participants may already be familiar with in the commercial. The textual explanation, which participants were instructed to locate and type into the text field, is highlighted with a pink stroke.

From section two onward, participants performed two tasks: first, identifying the most consumable explanation among the options presented (focusing on readability and accessibility), and second, selecting their preferred explanation design (focusing on personal aesthetic preference). This dual assessment allowed us to distinguish between objective usability and subjective preference.

This systematic evaluation of visual characteristics provided quantitative data on user preferences and perceptions, complementing the qualitative insights from the empathy workshop. Together, these two methodological phases (M01 and M02) provided a comprehensive understanding of the visual characteristics that influence the effectiveness of textual explanations in RecSys UIs, directly informing the development of the output dimension of f-RecX, as presented in Section 5.

4. Results

We present the analysis results of our two-phase research methodology (M01-M02) described in Section 3.

4.1. M01 — Empathy workshop: Qualitative and quantitative insights on commercial recommender systems

Our empathy workshop findings provide comprehensive answers to RQ1 and RQ2 across the four key objectives. The results for each objective are as follows.

Identification of Common RecSys Applications (Obj.1) | The results of the empathy workshop indicate that participants frequently interact with a diverse range of RecSys-driven applications. Table 1 presents the popularity of RecSys applications, structured by their respective domains. Streaming services, such as Spotify, Netflix, and YouTube, were the most frequently used (24 mentions, 38.7%), indicating their dominant role in shaping users' entertainment choices and an established use case of RecSys in non-experts' lives. Social media platforms followed closely, with 17 mentions, underscoring RecSys significant influence in personalizing content and connections in social networks. While streaming and social media applications led in usage, results also highlight the diversity of RecSys applications people engage with, including e-commerce (e.g., Zalando and Amazon) and health applications (e.g., Garmin and Apple Health).

Evaluation of Perceived Impact and Risk (Obj. 2) | We merged the different impact-risk coordinate systems of the four participant groups into one general coordinate system. The unified coordinate system again ranged from low (−1) to high (+1), for both impact and risk axes. Fig. 5 displays the results of this merge, using a convex hull

Table 1

Popularity ranking of preferred RecSys-driven applications among the participants of the empathy workshop (Obj.1). The results highlight the prevalence of streaming and social media applications as major RecSys-driven applications.

Domain	Count	Application
Streaming services	24	Spotify, Netflix, YouTube, Soundcloud, Audible
Social media	17	Instagram, Tinder, Facebook, Twitter
Other	6	Online database, Maps, Banking apps
E-commerce	6	Amazon, Zalando
Communication services	4	Mail, Outlook, Gmail
Health	3	Garmin, Google Health, Apple Health
News	2	Google News

technique to emphasize the spread of domains on the coordinate system. The mapping of the various RecSys-driven applications provided us with a valuable representation of users' impact-risk perceptions, shedding light on the possible RecSys-driven application domains that could benefit more from providing explanations.

The applications identified by participants were mainly distributed across three of the four quadrants, leaving the low-impact-high-risk quadrant nearly empty (only one entry point). This suggests that if a recommendation carries a high level of risk, it also tends to have a significant impact on users. This correlation between risk and impact has profound implications for the design of explainable Recsys. This suggests that explanations should be prioritized for high-impact domains rather than being implemented as a one-size-fits-all approach. This finding aligns with recent theoretical frameworks in XAI that emphasize context-sensitive explanation strategies (Suresh et al., 2021; Liao et al., 2020; Miller, 2019).

Streaming services and social media showed distinctly different risk-impact profiles. Streaming services were predominantly positioned in the lower quadrants, reflecting a low perceived risk, while the impact varied. Participants explained this by referring to the limited consequences of incorrect recommendations in entertainment contexts—at worst, users might miss preferred content or encounter inappropriate material. In contrast, social media applications were positioned in the upper-right quadrant (high impact, high risk). Participants raised concerns about the potential of these systems to influence opinions and behavior (Huertas, 2018; Voramontri and Klieb, 2019). Some participants expressed specific concerns that social media recommendation algorithms might lead users down “rabbit holes” of content, a phenomenon noted in previous research (Ledwich and Zaitsev, 2019; Harriger et al., 2022; Milton et al., 2023).

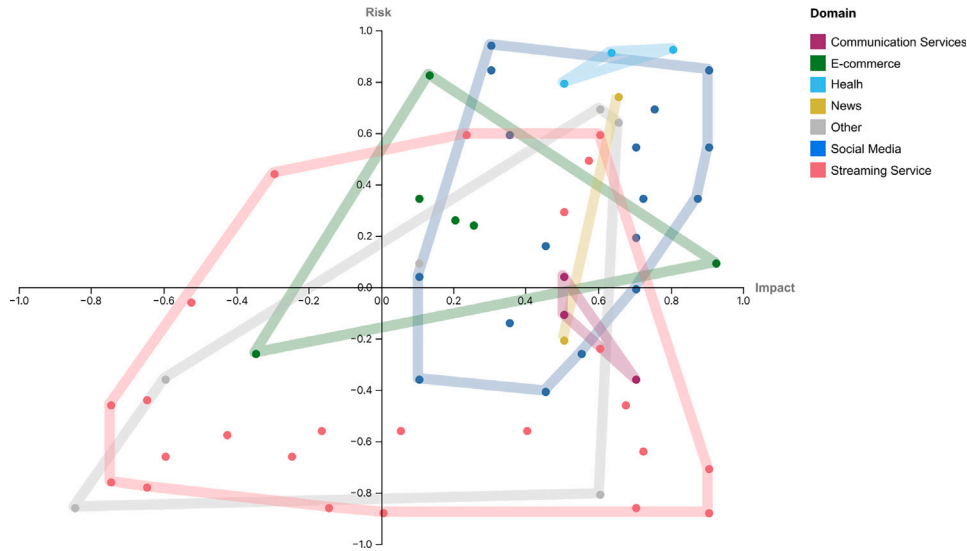


Fig. 5. Convex hull visualization of the results from M01 (Obj. 2) mapping the perceived impact (X-axis) and risks (Y-axis) associated with RecSys-driven applications (identified in Obj. 1). Each color represents a different domain, illustrating the spread and overlap in the risk-impact space. Social media and health applications are linked with substantial levels of both impact and risk (upper-right quadrant). In contrast, streaming services exhibit a low level of perceived risk but a diverse range of impact. Notably, the upper-left quadrant, representing low impact and high risk, remains almost unpopulated.

Table 2

Evaluation scores for five UI RecSys-driven applications across different domains, assessed by participants on four criteria: Correctly Located, Usefulness, Ease of Finding, and Satisfaction (Obj.3). The Amazon UI outperforms others with the highest scores in all criteria, resulting in the highest overall average score. The highest score in each category is highlighted in bold.

Application	Domain	Correctly located (%)	Usefulness	Ease of finding	Satisfaction	Average
Amazon	E-commerce	68.2	4.00	5.00	4.00	4.33
YouTube	Streaming service	63.1	3.35	3.53	3.23	3.37
Apple Music	Streaming service	68.2	3.00	4.00	2.50	3.17
Netflix	Streaming service	5.26 ^a	2.70	3.30	2.90	2.97
Spotify	Streaming service	63.1	2.60	2.40	2.40	2.80

^a The explanation provided by Netflix is divided into two sentences with only one participant being able to identify both, while the majority (68.2%) only located one part in the UI.

Quantitative Assessment of Explanations in RecSys UIs (Obj. 3)

| The assessment of the five commercial RecSys UIs provides insights into current layout strategies for textual explanations in terms of *ease of finding*, *satisfaction*, and *usefulness*. We measured these aspects using a 7-point Likert scale. Table 2 presents the results of participant evaluations across different applications.

On average, 66.16% of participants correctly located the explanations in the provided UIs. This indicates that in commercial RecSys UIs available to the majority of the population, one out of three explanations goes unnoticed. Amazon's RecSys UI received the highest scores across all categories, particularly excelling in "ease of finding" (5.00/7). This can be attributed to the consistent placement of textual explanations near the recommendation list and their visual prominence, achieved through bold, black font on a white background. When comparing music streaming services, Apple Music (3.17 average) outperformed Spotify (2.80 average), with the most notable difference in "ease of finding" (4.00 vs. 2.40). This suggests that Apple Music's larger font size, higher contrast, and better font weight make the explanation more accessible. The Netflix UI presents an interesting case with its two-part explanation displayed in different screen areas. The "We Think You'll Like" text at the top left was correctly located by 13/19 (68.2%) participants, while the "Because You Watched" text at the bottom was identified by only 1/19 (5.26%) participants. This suggests that separating explanations across different areas may hinder users' ability to fully understand the rationale behind recommendations. The perceived usefulness of textual explanations varies across different platforms, with Amazon scoring the highest at 4.00 and Spotify scoring the lowest at 2.60. These results indicate that, in an e-commerce context where users

must make a purchase decision, explanations are perceived as more useful than those in audio streaming services.

Several practical key factors influence the effectiveness of RecSys explanations: the proximity of the explanation to the recommendation, the visual clarity of the text, and the overall ease of identification. Improving these aspects could enhance user satisfaction and the overall utility of the recommendations provided by RecSys-driven applications.

User Preferences for RecSys Explanation UI Design (Obj. 4)

The moderated think-out-loud session demonstrated that an effective explanation requires a combination of both: high-quality content and effective presentation. Participant P4 highlighted this point: "Amazon's explanation relies on your previous behavior, which is evident because the explanation states that the recommended items are based on your previously viewed items. Moreover, I like that the explanation is placed just above the recommendation". This emphasizes the importance of both content and strategic placement for accessibility and relevance. P10 added, "I agree with you, but how can I identify which previous items the explanation refers to? It would also be beneficial to have this information so that I can better assess the new recommendations". This comment underscores that while users appreciate explanations based on past behavior, they desire clarity on which specific past actions influenced the recommendation. Thus, a more interactive or visually connected approach could enhance user comprehension and satisfaction.

Spotify and Netflix UIs received more critical feedback. Regarding Spotify UI P1, noted: "I never really paid attention to the sentence (i.e., explanation) that appears when I open a playlist. Typically, I simply click 'play' and then exit the application. Moreover, I usually

play music from my phone”. This reveals a disconnect between the explanation’s presentation and user engagement patterns, suggesting that the placement and timing of explanations may not align with user behavior and the device used. Participants consistently expressed preferences for *clear*, *easy to find*, and *accessible* explanations directly from the recommendation list. They preferred explanations to be prominently positioned at the top of the recommendation list to ensure good consumability. While participants did not express a strong preference for text-based or visually driven explanations, they emphasized the importance of interactivity as noted by P13, who expressed interest in interacting with explanations “as a way to know more about how and why” recommendations are made. This suggests that explanations should be designed as dynamic, interactive elements that encourage user engagement and exploration.

The empathy workshop revealed several key insights that informed the design of our subsequent online survey and the f-RecX framework.

Explanation Visibility: Two-thirds of participants could locate explanations in commercial UIs, indicating significant room for improvement in visibility.

Domain-Specific Risk-Impact: Different application domains carry varying levels of perceived risk and impact, with entertainment applications (low risk) and social media (high risk, high impact) occupying opposite ends of the spectrum.

Presentation Factors: Font size, contrast, weight, and positioning significantly influence explanation findability and perceived usefulness.

Placement Preferences: Users prefer explanations positioned near the recommendations they explain, ideally at the top of recommendation lists.

Interactive Elements: Users desire explanation interfaces that allow them to explore and understand the recommendation process beyond static text.

The findings on aspects like visibility, presentation factors, and placement directly informed the design of our online survey (M02). With M02 we systematically evaluated the visual characteristics that participants identified as most important for effective explanations.

4.2. M02 — Online user survey: Quantitative assessment of visual characteristics

Our systematic online survey of 38 participants provides quantitative evidence for RQ3 through six critical visual characteristics. We present the results for each visual characteristic below.

4.2.1. Familiarization task performance

In the familiarization task (see Fig. 4), 88% of the participants indicated that they found the textual explanation in the UI, with 84% correctly transcribing the text. Participants spent an average of 61.3 s (SD = 57.01) completing this task. This suggests that while most participants could identify its location within the UI, it was not immediately obvious to everyone and required a significant amount of time to locate.

4.2.2. Visual characteristic preferences

Table 3 presents a summary of the survey results, showing both the most consumable options (focusing on readability and accessibility) and participants’ preferred designs (focusing on aesthetic preference) for each visual characteristic.

Font size | 70.45% of participants identified the 32-pixel font size as the most consumable, indicating a clear preference for larger text in terms of readability. However, when it came to personal preference, 24 pixels emerged as the favored font size (36.3% of participants). This

discrepancy suggests that while larger fonts enhance readability, users prefer slightly smaller fonts that strike a balance between readability and efficient use of screen space. Both results align with accessibility guidelines for web design, which recommend a minimum font size of 16 px.

Font weight | ‘Bold’ font weight was rated as the most consumable by 34.09% of participants, with ‘Medium’ close behind at 32.82%. For personal preference, ‘Black’ was most favored (31.8%), followed by ‘Medium’ (27.3%). These findings suggest that while heavier font weights like ‘Bold’ and ‘Black’ are associated with improved consumability, there is some variation in aesthetic preference, with some participants favoring more moderate weights for visual balance.

Contrast | The most visually appealing combination of background color and text was a white background with black text (38.10% of participants), closely followed by the reverse contrast of a black background with white text (33.33%). Interestingly, the latter was much more preferred aesthetically (42.86% of participants). This suggests a potential trade-off between traditional standards and modern aesthetic preferences, where users may favor darker themes for visual comfort or stylistic reasons.

Placement of the explanation | The top-left placement, just above the recommendation list, was deemed both the most consumable (53.66%) and the most preferred placement (56.10%) for textual explanations. The center placement was the second most favored (36.59%). This strong preference for the top-left placement aligns with common Western UI design principles, where users typically begin scanning from the top-left corner of a screen (Jiang et al., 2023), reinforcing its effectiveness for displaying critical information, such as explanations.

Number of recommendation items | The majority of participants (62.50%) preferred seeing explanations alongside top-n recommendations, while only 5.0% favored a single recommendation, and the rest of the participants showed a preference for multiple recommendations ($m > n$, in our case, around 20 items). This finding suggests that users prefer a more comprehensive view that provides context through multiple options rather than focusing on a singular recommendation or a wall of recommendations, which may feel too restrictive or too informative in a decision-making process.

Length of the textual explanation | 45% of participants preferred an explanation length that nearly filled the entire width of the display (approximately 20 words). Notably, no participant favored explanations as long as a short paragraph. This preference for concise, visually balanced explanations suggests that while users value detailed information, they prefer it to be delivered in a concise format that does not overwhelm the interface or require excessive reading.

The survey results reveal important patterns that informed the development of f-RecX.

Visibility vs. Aesthetic Balance: Users consistently differentiate between what is most consumable (large fonts, high contrast) and what is most aesthetically pleasing (slightly more moderate sizes and weights).

Positioning Importance: The top-left position emerged as strongly preferred for explanations, suggesting this should be a default placement in recommender system UIs.

Context Through Multiple Items: Users prefer seeing explanations in the context of multiple recommendations rather than single items.

Conciseness Value: Brief but informative explanations are strongly preferred over lengthy, detailed ones, with medium-length explanations (around 20 words) being optimal.

Contrast Considerations: While traditional high contrast patterns (black on white) are seen as most consumable, modern inverse schemes (white on black) are aesthetically preferred.

Table 3
Results indicating the most consumable and most preferred visual characteristics of textual explanation and RecSys UI study in the online survey (M02).

Visual characteristic	Most consumable	Preferred design
Font size	32-pixel	24-pixel
Font weight	Bold	Black
Contrast	White background and black text	Black background and white text
Placement	Top-left close to the recommendation	Top-left close to the recommendation
Number of items	top-n	top-n
Length of the explanation	20 words	20 words

These empirical findings provide clear guidance for our framework’s visual representation dimension, offering evidence-based recommendations for designers seeking to optimize textual explanations in RecSys UIs.

5. f-RecX

f-RecX, our primary contribution, is a conceptual framework for characterizing and designing effective textual explanations in the UIs of RecSys. The key idea behind f-RecX is to guide designers and developers in creating effective textual explanations in RecSys UIs using a human-centered approach. f-RecX represents the first framework to integrate RecSys technical capabilities with visual design parameters specifically for textual explanations. Unlike existing frameworks that address either technical or visual aspects in isolation, f-RecX provides unified guidance across the complete design pipeline.

We conceptualized f-RecX as an interchangeable and expandable function that maps four input dimensions (in) to a single output dimension (out). Fig. 1 illustrates the complete framework. The input dimensions reflect different aspects of explainable RecSys: *Explanation Style* (in1), *Explanation Goals* (in2), *Domain Dynamics* (in3), and *Recommender Systems Technique* (in4). The output dimension is *Visual Representation of Recommendations and their Textual Explanation* (out1). Each dimension is characterized by either a spectrum or a set of categories that reflect different aspects of explainable RecSys, allowing for a fine-grained typological characterization. The dimensions and corresponding sub-dimensions of f-RecX are presented below, based on the results from the two-phase methodology described in Section 3 and in the literature. f-RecX serves as both a design tool for creating new explainable recommender systems and an evaluation framework for assessing existing implementations. The framework is demonstrated through usage scenarios (Section 6.3) and case study analyses (Section 6.4), showing its practical applicability across different domains and contexts.

5.1. Input dimensions

The four input dimensions of f-RecX address our first and second research questions (RQ1: “How do user preferences influence the comprehensibility and effectiveness of explanations in recommender systems?” and RQ2: “What specific visual characteristics of textual explanations most significantly impact user understanding?”) **Explanation Style** and **Explanation Goals** dimensions build upon taxonomies developed by Tintarev and Masthoff (2007, 2015), while incorporating insights from our empathy workshop (M01, Obj. 4) where participants expressed varying preferences for different explanation approaches. **Domain Dynamics** emerged directly from our impact-risk assessment activity (M01, Obj. 2), where participants’ mapping revealed how application context significantly influenced perceived explanation utility. This dimension addresses the contextual aspects of RQ1 by acknowledging that the effectiveness of explanations varies across domain boundaries. **Recommender Systems Technique** dimension reflects the underlying algorithmic approaches documented in comprehensive surveys (Zhang et al., 2020; Ko et al., 2022), recognizing that technical constraints influence what explanatory information is available for presentation to users.

5.1.1. Explanation style (in1)

This dimension refers to the approach used in the explanation to RecSys users and is characterized by a set of styles proposed by Tintarev and Masthoff (2012, 2007):

Case-Base style explanations explain the recommendations by comparing the situation of the current user with past cases.

Collaborative-Based style explanations are based on the preferences and behaviors of similar users.

Content-Based style explanations recommend items similar to those the user has previously interacted with, based on item attributes.

Demographic-Based style explanations use demographic information of users, such as age, gender, or location, to make and justify recommendations.

Knowledge-Based style explanations rely on domain knowledge about how certain item features meet user needs.

Utility-Based style explanations use a utility function to recommend items that maximize the user’s satisfaction.

Our empathy workshop findings (Section 3.1) revealed that users respond differently to different explanation styles. For example, in e-commerce settings such as Amazon, case-based explanations (“recommended because you viewed...”) were rated as most useful and satisfying (Table DD). This style leverages the familiarity of the user’s past experience to build trust in new recommendations. In contrast, our participants found content-based explanations particularly effective in entertainment domains, where recommendations based on attributes like genre or artist did not help them understand why items were suggested.

The **Explanation Style** dimension offers designers a strategic lens for crafting user interactions. **What:** This dimension refers to the approach used in the explanations. **Why:** The Explanation Style determines how information is framed and presented to users, affecting their understanding in the system. **How:** Designers can use this dimension to select the most appropriate Explanation Style based on the type of data available and the nature of the recommendations, ensuring explanations align with user expectations and system capabilities.

5.1.2. Explanation goals (in2)

This dimension identifies the objectives behind providing the explanation. These goals are not mutually exclusive and often work in combination to enhance the UX. This dimension incorporates the seven goals proposed by Tintarev and Masthoff (2007) and two additional

goals: ‘simplicity’ and ‘timeliness’ that emerged from our workshop findings 3.2.

Transparency explains how the RecSys works.

Scrutability allows users to tell the RecSys it is wrong.

Trust increases users’ confidence in the RecSys.

Effectiveness helps users make good decisions.

Persuasiveness convinces users of the recommended item.

Efficiency helps users make decisions faster.

Satisfaction increases ease of use or enjoyment.

Simplicity ensures the explanation complexity is easy to understand.

Timeliness provides prompt and immediate explanations.

Our workshop findings revealed that users especially value trust, effectiveness, and simplicity in explanations. For example, P4 highlighted the importance of trust by appreciating Amazon’s explanation of how recommendations are connected to past behavior. P13’s desire for interactive explanations reflects a need for transparency and scrutability in order to understand and potentially influence the recommendation process.

Different goals create different relationships between users and the RecSys. Prioritizing scrutability encourages user control and feedback loops, fostering human–AI collaboration. In contrast, focusing on effectiveness or efficiency optimizes decision support, which is particularly valuable in time-sensitive contexts or in high-risk, high-impact domains. As noted by workshop participants, simplicity and timeliness are crucial across all contexts. Explanations must be comprehensible and delivered at the right moment in the users’ journey.

The **Explanation Goals** dimension captures the fundamental purpose behind generating and presenting recommendation explanations, transforming technical outputs into user experiences.

What: The Explanation Goals dimension identifies the objectives behind providing the explanation, such as transparency, trust, effectiveness, and others.

Why: Clear goals are essential for creating purposeful explanations that enhance user experience and address specific user needs.

How: By prioritizing certain goals, designers can tailor explanations to achieve desired outcomes.

5.1.3. Domain dynamics (in3)

This dimension acknowledges external (human-dependent) factors that can influence the explanation design. It emphasizes the importance of considering the user’s context and needs to foster a human-centered approach to explainable RecSys. These factors are measured on a scale from 0 (low) to 1 (high) and include:

Impact refers to the degree of influence the RecSys has on users, ranging from low to high.

Risk refers to the potential negative outcomes of accepting a recommendation, such as privacy concerns or incorrect suggestions, ranging from low to high.

Urgency refers to how quickly the recommendation needs to be acted upon, with a spectrum from non-urgent to urgent.

Familiarity reflects the degree to which the user is familiar with the domain, ranging from unfamiliar to highly familiar.

The impact and risk factors were directly supported by our empathy workshop findings (Section 4.1), where participants mapped different RecSys applications along these dimensions. As shown in Fig. 5, streaming services were generally perceived as low-risk and variable-impact,

while social media recommendations were seen as both high-impact and high-risk. This variance in perception suggests that explanation strategies should adapt accordingly, providing more detailed and transparent explanations for high-risk contexts. Our findings also suggest that familiarity has a significant influence on how users interact with explanations. Workshop participants reported paying less attention to explanations in domains with which they were highly familiar (e.g., music streaming), while desiring more detailed explanations in less familiar contexts.

The **Domain Dynamics** dimension acknowledges the complex external factors that shape user interactions with recommender systems, recognizing that context is paramount in designing effective explanations.

What: The Domain Dynamics dimension considers external factors like impact, risk, urgency, and user familiarity in RecSys applications.

Why: Understanding the recommendation context is crucial for providing effective explanations that match the stakes involved and users’ knowledge levels.

How: Designers can adjust explanations based on associated risk or impact to ensure contextual appropriateness, providing more detailed explanations for high-risk or unfamiliar domains.

5.1.4. Recommender systems techniques (in4)

This dimension reflects the underlying algorithm techniques used to generate the recommendations. It connects to the technical foundation of the system and serves as a guideline for understanding what data can be used to generate explanations:

Content-Based, where recommendations are based on item features.

Context-Aware, where recommendations consider the operating context of the RecSys.

Collaborative Filtering, where recommendations are based on the preferences and behaviors of similar users.

The underlying technique influences what data is available for creating explanations. Content-based systems can leverage specific item attributes in explanations, while collaborative filtering systems can highlight community trends or similar users’ preferences. Workshop participants expressed particular interest in understanding which aspects of their behavior influenced recommendations, suggesting that explanations should connect to the underlying recommendation technique.

The **Recommender Systems Techniques** dimension bridges the gap between complex algorithmic processes and user-comprehensible explanations, revealing the technical mechanism behind recommendations.

What: The Recommender Systems Techniques dimension reflects the underlying method used to generate recommendations.

Why: The recommendation algorithm technique influences what data is available for explanations and how the system’s decision-making process can be explained.

How: This dimension guides designers in crafting explanations that accurately reflect the system’s functionality, ensuring explanations are consistent with how recommendations are generated.

5.2. Output dimension

Visual Representation of Recommendations and their Textual Explanation addresses our third research question (RQ3: “How can

the design of textual explanations be optimized for explainable recommender system user interfaces?”). This dimension synthesizes empirical findings from our quantitative survey of visual characteristics (M02).

5.3. Visual representation of recommendations and their textual explanation

This dimension focuses on how textual explanations and recommendations are presented to users. It encompasses the various visual characteristics studied in our online survey (Section 3.2) and consists of:

Font Size of the explanation, ranging from small (10 px) to large (32 px).

Font Weight of the explanation, from thin to black, affecting emphasis and visual hierarchy.

Contrast between the textual explanation and the background, ranging from a contrast ratio of -21 (no contrast) to $+21$ (high contrast).

Length of the explanation, ranging from a single word to a paragraph.

Position of the explanation on the UI.

Number of Recommendation Items, indicating whether recommendations are shown for single or multiple items.

Our survey results provided clear guidance for optimizing these characteristics. The font size should be at least 16 px, with 24–32 px preferred for better visibility. Bold or medium font weights enhance consumability while maintaining aesthetic appeal. High contrast (black on white or white on black) is essential for readability, and darker backgrounds are often preferred aesthetically. Explanations should be positioned in the top-left of the interface near the recommendations they explain. Multiple recommendation items (rather than single items) provide better context. Finally, explanations should be concise (around 20 words), avoiding lengthy paragraphs.

These visual aspects work together to create an effective visual hierarchy, enhancing the overall usability and UX of an explainable RecSys UI. The output dimension incorporates user preferences from our empirical studies and established accessibility guidelines to effectively guide designers and developers.

The **Visual Representation** dimension transforms abstract textual recommendation explanations into tangible, accessible visual elements that directly impact user understanding and engagement.

What: This output dimension focuses on how textual explanations and recommendations are visually presented to users.

Why: The visual presentation of explanations significantly affects their consumability, impacting how easily users can find, understand, and derive value from them.

How: Designers can use this dimension to optimize the visual characteristics of explanations, ensuring they are not only informative but also easily accessible and readable within the UI.

6. Evaluation

6.1. Comparison with existing XAI frameworks

To highlight the novel contributions of f-RecX, Table 4 presents a comparison with existing frameworks for designing UI for XAI systems. We evaluate the frameworks across the f-RecX dimensions: explanation style, explanation goal, domain dynamics, RecSys techniques, and visual representation and textual explanation.

This systematic mapping reveals substantial limitations in current research approaches that f-RecX specifically addresses. While *Explanation Goals* receive substantial attention across most frameworks, other crucial aspects like *Visual Representation and Textual Explanation* remain largely unexplored. *Domain Dynamics* and *Recommender System Techniques* are inconsistently addressed, with most frameworks providing limited coverage. The f-RecX framework distinguishes itself by comprehensively addressing all dimensions, offering an integrated approach to explainable recommender systems.

Our analysis reveals several patterns in existing frameworks that demonstrate the necessity for f-RecX:

Technical Integration: Most XAI frameworks treat technical aspects as secondary considerations, focusing primarily on high-level design principles without connecting them to the underlying algorithms’ capabilities and constraints. While Herlocker et al. (2004) established that collaborative filtering explanations enhance persuasiveness, current UI frameworks fail to connect this finding with concrete visual design parameters. f-RecX addresses this by explicitly linking collaborative filtering capabilities (highlighting similar users’ preferences) with specific visual representation guidelines (font-weight for emphasis, positioning for discoverability). Moreover, framework like Chromik and Butz (2021) provides extensive theoretical guidance but offer no connection between content-based recommendation techniques and appropriate explanation styles. For instance, content-based systems can leverage item feature explanations (“recommended because of the genre: sci-fi”), but existing frameworks do not specify how to visually present these feature-based explanations or optimize their textual format for different user contexts. **Human-Interaction Focused Frameworks:** A significant category of XAI frameworks places primary emphasis on understanding and optimizing human–AI interaction patterns, with particular attention to how users mentally process and respond to algorithmic explanations. Chromik and Butz (2021) and Chromik and Schuessler (2020) prioritize interaction concepts and human evaluation methodologies. While they thoroughly address explanation goals, they offer minimal guidance on visual representation, precisely where f-RecX provides the concrete design parameters essential for effective implementation. This illustrates the tendency to emphasize theoretical components while neglecting the practical visual elements that determine whether users consume explanations. Moreover, these frameworks excel in defining interaction paradigms and evaluation metrics but fail to bridge the gap between user needs and technical implementation. For instance, while Chromik and Butz (2021) identify trust-building as a key explanation goal, they do not specify how collaborative filtering systems should present trust-building explanations (such as highlighting consensus among similar users) in terms of specific visual design parameters like font hierarchy, color coding, or positioning strategies that f-RecX explicitly addresses.

User-Centric Process: Another prominent approach in explainable AI research emphasizes the methodological aspects of involving users throughout the design and development process, recognizing that effective explanations must be grounded in actual user needs rather than designer assumptions. Förster et al. (2020), Eiband et al. (2018), and Nguyen et al. (2024) focus on process-oriented approaches. Despite their valuable contributions to human-centered design processes, they focus less on specific visual implementation guidance and fail to connect these processes with the technical realities of recommender systems—a critical integration that f-RecX uniquely provides. While these frameworks provide excellent methodological foundations for involving users in the design process, they lack concrete guidance on translating user feedback into specific visual design decisions. Moreover, frameworks like Eiband et al. (2018)’s participatory approach provide valuable insights into user mental models, but do not connect these insights with the capabilities of recommendation algorithms. For instance, when users express preference for case-based explanations (“because you liked X”), existing frameworks do not specify

how to visually represent these explanations in content-based systems where similar items are determined by feature similarity rather than user behavior patterns—a technical-visual integration that f-RecX systematically addresses.

Stakeholder and Goal-Oriented Frameworks: A well-established tradition in XAI research focuses on systematically analyzing the diverse needs and objectives of different stakeholders who interact with AI systems, recognizing that explanation requirements vary significantly across user roles, expertise levels, and decision-making contexts. Suresh et al. (2021), Wang et al. (2019), and Liao and Varshney (2021) excel in analyzing explanation goals and stakeholder needs. However, they leave practitioners without actionable guidance on translating these insights into effective visual implementations. f-RecX directly addresses this gap through empirically validated visual design parameters. While these frameworks provide comprehensive taxonomies of explanation goals and stakeholder requirements, they lack the bridge between conceptual understanding and practical implementation. Moreover, frameworks like Suresh et al. (2021)'s multi-level approach effectively map objectives to tasks but do not specify how different RecSys techniques should visually represent these tasks. For instance, when addressing the goal of building user trust through transparency, existing frameworks do not distinguish how collaborative filtering systems should visually present trust-building information (showing similar users' preferences) versus content-based systems (highlighting item feature alignments), leaving practitioners without algorithm-specific visual guidance that f-RecX provides through its integrated approach.

Comprehensive but Domain-Agnostic Frameworks: Some frameworks attempt to provide holistic coverage of explainable AI principles by developing general-purpose approaches that can be applied across multiple domains and application contexts, prioritizing broad applicability over domain-specific optimization. The EUCA framework (Jin et al., 2021) and Herm et al.'s design theory (Herm et al., 2022) offer more balanced coverage but remain domain-agnostic. Their general applicability comes at the cost of specific guidance for RecSys, where f-RecX provides the domain-specific parameters that practitioners require for effective implementation. While these frameworks provide valuable overarching principles that apply across various AI domains, they lack the specificity needed for effective RecSys explanation design. Moreover, frameworks like EUCA offer excellent theoretical foundations but do not address the unique challenges of recommendation explanation visualization. For instance, while EUCA provides general principles for explanation clarity and user comprehension, it does not specify how to visually differentiate between collaborative filtering explanations (requiring emphasis on user similarity) and content-based explanations (requiring emphasis on item features), nor does it provide concrete visual parameters like font weights, positioning strategies, or color schemes optimized for recommendation contexts—domain-specific guidance that f-RecX delivers through its empirically validated approach.

The current landscape features frameworks that either address theoretical aspects without practical implementation guidance or provide general principles without recommender-specific parameters. By integrating RecSys's technical aspects with concrete human-centered design guidelines and empirically validated visual parameters, f-RecX fills a critical gap.

For researchers and practitioners seeking to implement effective textual explanations in RecSys, f-RecX offers unprecedented value by recognizing that the technical foundation determines explanation possibilities, and visual design must align with these technical realities to create truly effective explainable systems. The framework ensures that explanations of design decisions are grounded in both algorithmic capabilities and human-centered design principles, creating a bridge that existing frameworks consistently fail to provide.

While this comparison reveals f-RecX's unique comprehensive coverage, we acknowledge that our analysis is primarily descriptive rather than empirical. A rigorous validation would require controlled experiments where the same explainable RecSys interfaces are designed using different frameworks and compared using standardized metrics.

6.2. Illustrative examples

We evaluate f-RecX by applying it to existing interfaces to illustrate its applicability across RecSys UIs. In Section 6.3, we present a usage scenario demonstrating how designers can apply the framework when creating a new system. In Section 6.4, we analyze two existing RecSys UIs through the f-RecX lens, assessing their alignment with our framework principles. This dual approach highlights the flexibility of the framework in both the design and assessment phases of RecSys development.

6.3. Usage scenario

To demonstrate the descriptive power (Beaudouin-Lafon, 2004) and the practical application of f-RecX in the design process, we use a persona-based scenario (Haag et al., 2007), describing the ideal experience from a specific user's perspective. This method provides a clear understanding of how the framework can be effectively utilized in real-world contexts.

Consider a team of RecSys developers and UI designers working on a new music streaming platform called MelodyMix. The platform aims to provide users with both personalized song recommendations and textual explanations to improve overall user engagement. To achieve this goal, the MelodyMix team leverages the f-RecX framework to guide the design of explanations that align with user needs, adhere to RecSys explainability guidelines, and meet accessibility standards.

The MelodyMix team identifies three primary user personas: *casual listeners*, *music enthusiasts*, and *social discoverers*. Casual listeners primarily stream music during commutes, workouts, or while engaging in other background activities. They prefer quick, effortless discovery without detailed explanations. Music enthusiasts actively explore new artists and genres. They appreciate understanding why specific songs are recommended and value discovering musical connections. Lastly, social discoverers enjoy music through social connections and community trends. They prefer explanations that highlight what others with similar tastes are enjoying.

The team begins by considering the input dimensions of the f-RecX framework.

Explanation Style (in1): After analyzing user demographics and available data sources, the team decides on a combination of *Collaborative-Based* and *Content-Based* styles. Collaborative-based explanations will allow users to understand how music discoveries are driven by community trends. Content-based explanations will enable users to explore how the recommended songs share audio features, e.g., tempo and genre, with previously liked songs.

Explanation Goals (in2): Informed by f-RecX, the team prioritizes *Trust* and *Satisfaction* as primary goals, aiming to build user confidence by transparently showing how recommendations are generated and enhancing the overall UX through clear, helpful explanations.

Domain Dynamics (in3): For a music streaming service, the team assesses that the *Impact and Risk* are both relatively low (as supported by our workshop findings in Section 4.1). The *Urgency* is considered medium, as some users want to quickly decide what to listen to next, while others may browse more leisurely. They estimate a medium level of *Familiarity* among their target users, considering both music enthusiasts and casual listeners.

Recommender Systems Technique (in4): Given the nature of the music platform, the team implements a *hybrid* system. This aligns with their chosen explanation styles, allowing them to provide explanations that highlight both specific song attributes and community preferences.

Table 4

Detailed comparison of XAI frameworks. Each row represents a specific aspect within the six main dimensions of f-RecX. Legend: ■ = Addressed with specific parameters; ■ = Partially addressed with some guidance; □ = Not addressed.

Framework aspects		f-RecX	Chromik and Butz (2021)	Chromik and Schuessler (2020)	Nguyen et al. (2024)	Suresh et al. (2021)	Förster et al. (2020)	Liao et al. (2020)	Liao and Varshney (2021)	Jin et al. (2021)	Herm et al. (2022)	Eiband et al. (2018)	Wang et al. (2019)
Explanation style	Content-based	■	■	□	□	□	□	■	■	■	■	□	□
	Collaborative	■	■	□	□	□	□	□	■	■	■	□	□
	Case-based	■	■	□	□	□	□	□	■	■	■	□	□
	Demographic-based	■	□	□	□	□	□	□	□	■	□	□	□
	Knowledge-based	■	■	□	□	□	□	□	■	■	■	□	□
	Utility-based	■	■	□	□	□	□	□	■	■	■	□	□
Explanation goals	Trust	■	■	■	■	■	■	■	■	■	■	■	■
	Transparency	■	■	■	□	■	■	■	□	■	■	■	■
	Effectiveness	■	■	■	□	■	■	■	□	■	■	□	□
	Persuasiveness	■	■	■	□	□	■	□	□	□	□	□	□
	Satisfaction	■	■	■	□	■	■	□	□	■	□	□	□
	Scrutability	■	■	■	□	□	□	□	□	■	□	□	□
	Efficiency	■	□	□	□	□	□	□	□	■	□	□	□
	Simplicity	■	□	□	□	□	□	□	□	□	□	□	□
	Timeliness	■	□	□	□	□	□	□	□	□	□	□	□
Domain dynamics	Impact	■	□	■	■	□	□	□	■	■	□	■	□
	Risk	■	□	□	□	□	□	□	■	■	□	□	□
	Urgency	■	□	□	■	□	□	□	■	■	□	□	□
	Familiarity	■	□	□	■	□	□	□	■	■	□	■	□
RecSys techniques	Content-based	■	■	□	□	□	□	□	□	■	■	□	■
	Collaborative filtering	■	□	□	□	□	□	□	□	□	□	□	□
	Context-aware	■	□	□	□	□	□	□	□	□	□	□	□
Visual representation	Font size	■	■	□	■	■	□	□	□	■	□	□	□
	Font weight	■	■	□	■	■	□	□	□	■	□	□	□
	Contrast	■	□	□	□	□	□	□	□	□	□	□	□
	Position	■	■	□	□	■	□	□	□	■	□	□	□
	Number of recommendations	■	□	□	□	□	□	□	□	□	□	□	□
	Length	■	□	□	□	□	□	□	□	■	□	□	□

Based on these input dimension decisions, the team then focuses on the output dimension, guided by our empirical findings.

Font Size: Following the survey results showing user preference for 24 px font (Section 4.2), the team implements this size for the explanation text, ensuring good visibility while maintaining aesthetic balance.

Font Weight: The team selects a medium font weight for explanations, balancing readability with visual hierarchy within the UI.

Contrast: Based on user aesthetic preferences for dark themes (Section 4.2), they implement white text on a dark background with a high contrast ratio meeting accessibility standards. **Position:** Following the strong user preference for top-left positioning (53.66% in our survey), explanations are placed directly above recommendation lists. **Number of Recommendation Items:** Implementing the preferred “top-n” approach (favored by 62.50% of survey participants), the team displays explanations alongside a carousel of five recommended songs. **Length:** Adhering to user preferences for concise explanations (Section 4.2), they limit explanation text to approximately 20 words, providing sufficient context without overwhelming the interface.

Fig. 6 illustrates two possible resulting MelodyMix interface designs, demonstrating the practical application of f-RecX. The interfaces showcase how systematic design decisions translate f-RecX specifications into user-centered explanations: *collaborative-based* explanations (“Fans of [Artist] also enjoyed these tracks”) are positioned at the top-left following the 53.66% user preference from our survey, while *content-based* explanations (“Similar tempo and energy to [Your Liked Song]”) appear beneath each recommendation, both using 24 px font size and

medium weight on high-contrast dark backgrounds as specified by the framework. Trust is fostered through transparent reasoning that references users’ actual listening history (e.g., “you loved in ‘505’” or “like ‘Brianstorm’ you loved”), making the recommendation logic explicit and personally verifiable. Satisfaction is achieved by avoiding technical jargon in favor of accessible language (“high energy indie rock” rather than algorithmic terminology) and employing personalized phrasing (“your taste”, “your vibe”) that emphasizes relevance to individual users. The 20-word explanation limit ensures cognitive accessibility while the consistent visual hierarchy guides users intuitively from explanation to action. This concrete instantiation demonstrates f-RecX’s ability to systematically translate user preferences and explainability goals into coherent interface designs that can be empirically evaluated for their effectiveness in building user trust and satisfaction with recommender systems.

6.4. Mapping of existing RecSys UIs

To demonstrate the descriptive power (Beaudouin-Lafon, 2004) of f-RecX, the second application is illustrated via a *case study analysis* two contrasting RecSys UIs. We demonstrate how f-RecX can be applied to assess the human-centered aspects of existing systems, utilizing both commercial and academic RecSys UIs. These systems are particularly suitable for analysis because they are publicly available, represent different domain contexts with varying risk profiles, and both adopt textual explanations that can be systematically evaluated using our

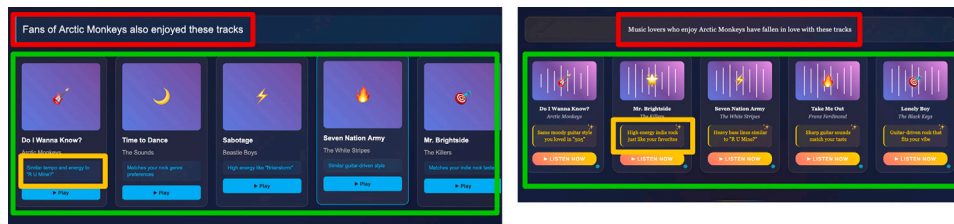


Fig. 6. Comparison of explainable music recommender UI designs following the f-RecX framework. Both implementations feature collaborative-based explanations positioned at top-left (red), content-based explanations under each song (yellow), and top-5 recommendation display (green). All designs maintain 24 px explanation text, high contrast, and 20-word explanation limits as specified by f-RecX guidelines.

framework. We deliberately selected two cases that represent opposite ends of the risk-impact spectrum identified in our empathy workshop findings. This approach enables us to demonstrate f-RecX's applicability across various domain dynamics, while providing both validation of successful implementations and identification of opportunities for improvement in systems that may not fully align with user preferences. The UIs of these two applications are visible in Figs. 7 and 8.

Case 1: Apple TV RecSys UI – It represents a low-risk, high-familiarity entertainment context where recommendation errors have minimal negative consequences. This case serves to validate f-RecX against a successful commercial implementation that millions of users interact with daily. Our objective was to assess the effectiveness of the textual explanation, “Enjoy personalized recommendations based on your past plays” visible in Fig. 7. The textual statement can be interpreted as serving the ‘trust’ Explanation Goals, by building user confidence, and clarifying that the recommendations are tailored to users’ previous viewing activity. Moreover, it can be classified as a ‘case-based’ Explanation Style as the explanation refers to past user behavior. In examining the Domain Dynamics, we identified that the Apple TV RecSys operates within a context of ‘low’ impact, ‘low’ risk, and ‘low’ urgency, with users generally possessing a ‘medium to high level’ of familiarity within the movie domain. We hypothesized that the RecSys Techniques employed by Apple TV is content-based, though this remains speculative due to the system’s black-box nature. The interface’s design choices, particularly in terms of Visual Representation of Recommendations and their Textual Explanation, play a crucial role in reinforcing user trust. Specifically, the textual explanation is presented in a medium-weight font with a size between 16 px and 18 px, achieving a contrast ratio of 3.95:1. While this ratio is only marginally acceptable, the large font size ensures readability and complies with accessibility rules. The message is short, consisting of just eight words, and is strategically positioned at the top left of the screen, directly above the recommendation carousel. This placement ensures that the explanation is immediately visible to the user, guiding their attention toward the carousel, which displays multiple personalized recommendations in a visually engaging format. Overall, the Apple TV RecSys UI does a good job of conveying the explanation to users, as it is easily localizable and consumable.

Case 2: Health RecSys UI – It represents a high-risk, medium-familiarity health context where recommendation decisions can have significant real-world consequences. This case demonstrates f-RecX's utility in identifying critical design gaps in high-stakes applications. In this case (Bhattacharya et al., 2023), we evaluated the effectiveness of textual explanations in the marked box VC4 in Fig. 8. Each recommendation in this section is accompanied by an explanation stating “Other patients like Patient - ### who have taken this action have [EFFECT] risk”. This explanation employs a ‘collaborative-based’ Explanation Style as it references the behavior of similar users. Regarding Explanation Goals, the explanation serves both ‘effectiveness’ and ‘persuasiveness’ functions. Effectiveness by providing evidence-based outcomes and persuasiveness by indicating the

positive results experienced by similar users, potentially motivating adoption of the same recommendation. For Domain Dynamics, we can infer that users of this health RecSys likely have a medium level of ‘familiarity’ with the domain. The ‘urgency’ appears high, as the interface contextualizes recommendations by associating them with concerning health metrics such as low physical activity and increased waist measurement. The levels of ‘impact’ and ‘risk’ can be classified as medium to high, reflecting the health consequences of following or ignoring the recommendations. Based on the explanation’s leverage of similar users’ behaviors, we can reasonably conclude that the RecSys Techniques implemented is collaborative filtering. Regarding

Visual Representation of Recommendations and their Textual

Explanation, the font size ranges between 10–12 px with mixed regular and medium font weights, creating a contrast ratio of 3.66:1. This combination fails to meet accessibility guidelines and does not align with our empirical findings on user preferences. However, the explanation length is appropriate at approximately 12 words, providing sufficient context about the recommendation while remaining concise. Although the explanation is positioned below the recommendation, contrary to our user study findings, which favor top-left placement, this arrangement does facilitate a direct association between the recommendation and its justification. Overall, this health RecSys UI effectively conveys explanations to users, with notable room for improvement in visual presentation aspects.

The case study analysis demonstrates f-RecX's practical application and descriptive power across diverse implementation contexts. The framework successfully identified optimization opportunities in a well-designed commercial system while revealing critical accessibility gaps in an academic research implementation. Most importantly, f-RecX can be used to suggest improvement rather than general recommendations, enabling designers to make evidence-based decisions about explanation interface design. This evaluation approach provides researchers and practitioners with a structured method for assessing and enhancing textual explanations in RecSys UIs across various domains, ensuring that explanation design aligns with both user preferences and accessibility requirements while effectively addressing the specific risk and impact characteristics of each application context.

7. Discussion and future work

We discuss the limitations of our study, potential future research directions, and broader implications for the field of explainable RecSys. Specifically, we discuss its alignment with prior work, critically examine our methodology, analysis results, and the proposed f-RecX framework, drawing connections to human-centered concepts and human-AI collaboration.

Alignment with Prior Research: Our findings confirm several established insights from previous research, particularly (Herlocker et al., 2004) observation that users appreciate explanations in collaborative filtering systems and Tintarev and Masthoff (2015) emphasis on explanation clarity and brevity. However, our research significantly extends prior work by providing the first investigation of visual presentation

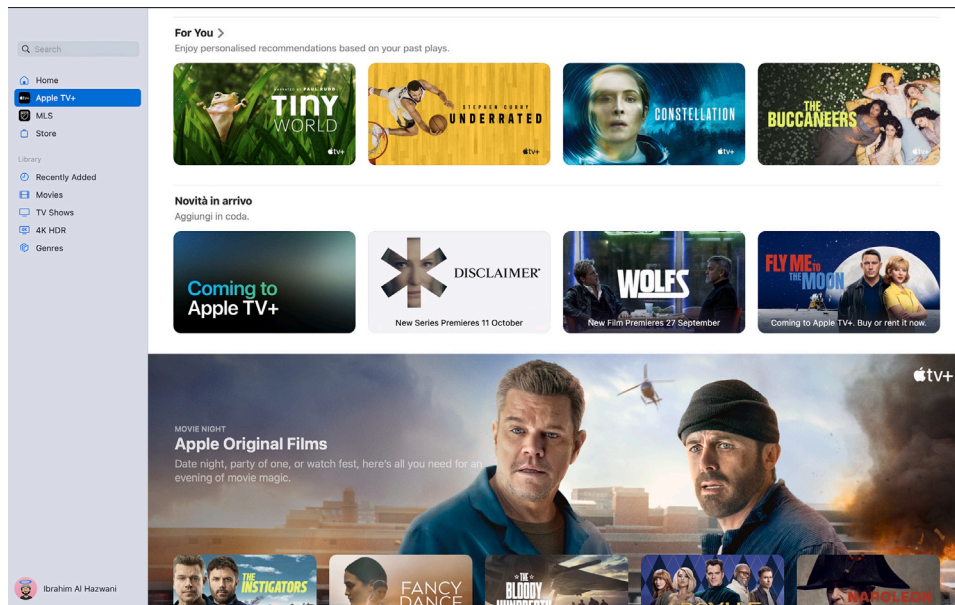


Fig. 7. Apple TV RecSys UI: The ‘For You’ section offers personalized movie and TV series recommendations. The text ‘Enjoy personalized recommendations based on your past plays’ under the header enhances trust (explanation goal) in the RecSys. Mapping this UI within the f-RecX framework reveals that the textual explanation aligns with users’ preferences for this class of RecSys.

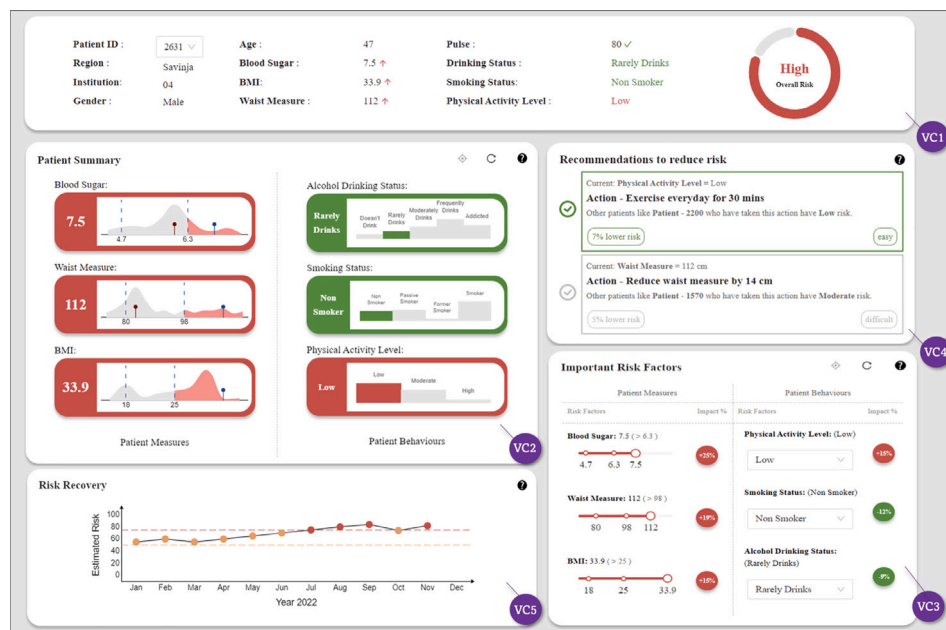


Fig. 8. RecSys UI from the work of Bhattacharya et al. (2023). The UI presents the user with multiple explanations, both visual and textual. The focus of our analysis was the group VC4, where the user is prompted with a recommendation to reduce the risk.

factors in explainable RecSys. While previous studies focused primarily on explanation content and algorithmic approaches, our research addresses the gap regarding the neglect of UI presentation factors in explanation design (Zhang et al., 2020; Ko et al., 2022; Chatti et al., 2024). Unlike existing frameworks that address either technical or visual aspects in isolation, f-RecX provides unified guidance across the entire design pipeline, introducing concepts such as explanation “consumability” and domain dynamics mapping that were absent in prior frameworks. Our validation of specific visual parameters provides design guidance that bridges the gap between theoretical insights and practical implementation in explainable RecSys.

Expand Device Dependency: Our study focused on desktop interfaces for RecSys explanations, providing a solid foundation for understanding user interactions. However, this focus limits the generalizability of our findings to other device contexts. As noted by Polatidis and Georgiadis (2013) and Ricci (2010), mobile devices present unique challenges and opportunities due to their smaller screens and touch-based interaction. Future research should investigate how textual explanations can be adapted for devices with varying screen sizes and interaction modalities. A cross-platform investigation examining how explanation preferences vary across devices would be particularly valuable for companies developing multi-device recommender systems.

Broadening User Demographics: While we deliberately focused on non-expert users to provide insights into the needs and preferences of the largest user group, this approach has limitations. Future studies should investigate how different user demographics interact with and interpret explanations. For example, comparing explanation preferences across age groups, technical expertise levels, and cultural backgrounds could yield important insights for creating more inclusive explanation designs. Additionally, investigating how expert users, such as data scientists or RecSys model builders, might benefit from more technical explanations would enable the design of adaptable interfaces that serve users with varying levels of domain knowledge.

Empirical Framework Validation: While our framework has been validated through user studies and case analysis, rigorous empirical validation represents a critical next step. Future research should implement controlled experiments comparing f-RecX-guided explanation designs against interfaces designed using existing frameworks such as EUCA (Jin et al., 2021), Chromik and Schuessler (2020), Chromik and Butz (2021), and baseline conditions. Such studies should employ standardized metrics including task performance (completion time, error rates), explanation effectiveness (findability, comprehension), and user experience measures (satisfaction, trust, perceived usefulness). Additionally, longitudinal studies of real-world implementations would provide valuable insights into the practical impact of f-RecX-guided explanation designs. A/B testing different explanation designs based on f-RecX dimensions could provide quantitative validation of the framework's effectiveness and establish its empirical superiority over alternative approaches.

Designing for Explanation 'Consumability': Our findings highlight the concept of explanation 'consumability'—how easily users can locate, understand, and derive value from explanations. This concept extends beyond traditional measures of explanation quality, which focus primarily on content accuracy and completeness. The visual characteristics we identified as important (font size, weight, position, etc.) significantly impact this consumability, suggesting that designers should consider these factors as essential components of explanation design rather than secondary styling concerns. This perspective aligns with recent calls in the literature for more holistic approaches to explanation design that consider both content and presentation (Eiband et al., 2018).

Integration with LLM-Generated Explanations: With the increasing adoption of LLMs for generating explanation content (Castelnovo et al., 2024; Silva et al., 2024; Sun et al., 2024), our framework offers timely guidance on how to effectively present this content. As LLMs enable more sophisticated, personalized, and natural language explanations, the presentation aspects we have identified become even more crucial for ensuring that users effectively consume these advanced explanations. Future research should specifically investigate how f-RecX principles can be adapted for different types of LLM-generated explanations. This could potentially lead to automated systems that generate explanation content and optimize their visual presentation based on context and user preferences.

Advancing Human-AI Collaboration: An exciting area for future research lies in exploring forms of human-AI collaboration for generating, consuming, and refining explanations in RecSys. Building on mixed-initiative methodologies (Abdul et al., 2018; Holter and El-Assady, 2024), future studies could investigate how users can actively shape explanations by guiding or refining AI-generated content. This shift from passive consumption to interactive, user-driven tools could foster deeper understanding and trust in recommendation systems. For example, systems might allow users to specify which aspects of recommendations they want explained or enable them to provide feedback that shapes future explanations.

In conclusion, our study has laid a comprehensive foundation for understanding user preferences in explainable RecSys. The proposed f-RecX framework offers a versatile approach to designing effective textual explanations in RecSys UIs. By pursuing the outlined future research directions, we can continue to advance the field, creating a more transparent, trustworthy, and user-centered RecSys.

8. Conclusion

Our study has laid the foundation for understanding user preferences in explainable Recommender Systems. The proposed conceptual framework, f-RecX, is the result of a two-step methodology that combines qualitative user workshops and quantitative visual preference assessments. It consists of four input dimensions (addressing explanation content, purpose, context, and algorithmic foundation) and one output dimension focused on visual representation, collectively forming a comprehensive design guide. f-RecX offers a versatile approach to designing effective textual explanations in RecSys UIs. It integrates technical aspects of recommendation generation with human-centered design principles and specific visual guidelines. By building on the limitations, we have acknowledged and are pursuing the outlined future research directions. The field can continue to advance, creating more transparent, trustworthy, and user-centered RecSys that effectively support decision-making across diverse domains and contexts. As LLMs and other AI technologies continue to enhance the generation of explanatory content, frameworks like f-RecX provide preliminary guidance for presenting this content effectively. However, rigorous empirical validation through controlled experiments remains necessary to establish their superiority over existing approaches and ensure maximum value to users.

CRedit authorship contribution statement

Ibrahim Al-Hazwani: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Gabriela Morgenshtern:** Writing – review & editing, Writing – original draft. **Mennatallah El-Assady:** Writing – review & editing, Supervision, Conceptualization. **Jürgen Bernard:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- Abdul, A., Vermeulen, J., Wang, D., Lim, B.Y., Kankanhalli, M., 2018. Trends and trajectories for explainable, accountable and intelligible systems: An HCI research agenda. In: ACM Conference on Human Factors in Computing Systems. CHI, pp. 1–18.
- Adadi, A., Berrada, M., 2018. Peeking inside the black-box: a survey on explainable artificial intelligence (XAI). *IEEE Access* 6, 52138–52160.
- Al-Hazwani, I., Alahmadi, T., Wardatzky, K., Inel, O., El-Assady, M., Bernard, J., 2021. Interaction tasks for explainable recommender systems. *Eurographics Conf. Vis. (EuroVis)* 37–39.
- Beaudouin-Lafon, M., 2004. Designing interaction, not interfaces. In: Costabile, M.F. (Ed.), *Advanced Visual Interfaces AVI*. ACM Press, pp. 15–22. <http://dx.doi.org/10.1145/989863.989865>.
- Beel, J., Dixon, H., 2021. The 'unreasonable' effectiveness of graphical user interfaces for recommender systems. In: ACM Conference on User Modeling, Adaptation and Personalization. UMAP, pp. 22–28.

- Bhattacharya, A., Ooge, J., Stiglic, G., Verbert, K., 2023. Directive explanations for monitoring the risk of diabetes onset: introducing directive data-centric explanations and combinations to support what-if explorations. In: *ACM Conference on Intelligent User Interfaces*. IUI, pp. 204–219.
- Biega, A.J., Potash, P., Daumé, H., Diaz, F., Finck, M., 2020. Operationalizing the legal principle of data minimization for personalization. In: *ACM Conference on Research and Development in Information Retrieval*. SIGIR, pp. 399–408.
- Cardoso, B., Sedrakyan, G., Gutiérrez, F., Parra, D., Brusilovsky, P., Verbert, K., 2019. IntersectionExplorer, a multi-perspective approach for exploring recommendations. *Int. J. Hum.-Comput. Stud.* 121, 73–92.
- Castelnovo, A., Depalmas, R., Mercorio, F., Mombelli, N., Poterti, D., Serino, A., Seveso, A., Sorrentino, S., Viola, L., 2024. Augmenting XAI with LLMs: A case study in banking marketing recommendation. In: *World Conference on Explainable Artificial Intelligence*. Springer, pp. 211–229.
- Chatti, M.A., Guesmi, M., Muslim, A., 2024. Visualization for recommendation explainability: a survey and new perspectives. *ACM Trans. Interact. Intell. Syst. (TIIS)* 14 (3), 1–40.
- Chen, X., Zhang, Y., Wen, J.-R., 2022. Measuring "why" in recommender systems: A comprehensive survey on the evaluation of explainable recommendation. *arXiv preprint arXiv:2202.06466*.
- Chromik, M., Butz, A., 2021. Human-XAI interaction: a review and design principles for explanation user interfaces. In: *IFIP International Conference on Human-Computer Interaction*. INTERACT, Springer, pp. 619–640.
- Chromik, M., Schuessler, M., 2020. A taxonomy for human subject evaluation of black-box explanations in XAI. *Exss-Atec@ Iui* 1, 1–7.
- Colin, J., Fel, T., Cadène, R., Serre, T., 2022. What i cannot predict, i do not understand: A human-centered evaluation framework for explainability methods. *Conf. Neural Inf. Process. Syst. (NeurIPS)* 35, 2832–2845.
- Confalonieri, R., Coda, L., Wagner, B., Besold, T.R., 2021. A historical perspective of explainable artificial intelligence. *Wiley Interdiscip. Rev.: Data Min. Knowl. Discov.* 11 (1), e1391.
- Das, A., Rad, P., 2020. Opportunities and challenges in explainable artificial intelligence (xai): A survey. *arXiv preprint arXiv:2006.11371*.
- Drouet, L., Sleeswijk Visser, F., Lallemand, C., 2023. Using empathy-centric design in industry: Reflections from the UX researcher, the client, and the method expert. In: *Empathy-Centric Design Workshop*. pp. 1–9.
- Eiband, M., Schneider, H., Bilandzic, M., Fazekas-Con, J., Haug, M., Hussmann, H., 2018. Bringing transparency design into practice. In: *ACM Conference on Intelligent User Interfaces*. IUI, pp. 211–223.
- Förster, M., Klier, M., Kluge, K., Sigler, I., 2020. Fostering human agency: A process for the design of user-centric XAI systems.
- Gedikli, F., Jannach, D., Ge, M., 2014. How should I explain? A comparison of different explanation types for recommender systems. *Int. J. Hum.-Comput. Stud.* 72 (4), 367–382.
- Ghajargar, M., Longo, L., Gargiulo, E., Giannantonio, R., 2017. Empathy workshop: When project team and pilot users exchange experiences. *Des. J.* 20 (sup1), S3837–S3848.
- Guesmi, M., Chatti, M.A., Joarder, S., Ain, Q.U., Alatrash, R., Siepmann, C., Vahidi, T., 2024. Interactive explanation with varying level of details in an explainable scientific literature recommender system. *Int. J. Hum.-Comput. Interact.* 40 (22), 7248–7269.
- Gunning, D., Stefik, M., Choi, J., Miller, T., Stumpf, S., Yang, G.-Z., 2019. XAI—Explainable artificial intelligence. *Sci. Robot.* 4 (37), eaay7120.
- Haag, G., Kuhlman, R., Mante, D., 2007. About face 3: the essentials of interaction design.
- Harriger, J.A., Evans, J.A., Thompson, J.K., Tykka, T.L., 2022. The dangers of the rabbit hole: Reflections on social media as a portal into a distorted world of edited bodies and eating disorder risk and the role of algorithms. *Body Image* 41, 292–297.
- He, C., Parra, D., Verbert, K., 2016. Interactive recommender systems: A survey of the state of the art and future research challenges and opportunities. *Expert Syst. Appl.* 56, 9–27.
- Herlocker, J.L., Konstan, J.A., Terveen, L.G., Riedl, J.T., 2004. Evaluating collaborative filtering recommender systems. *ACM Trans. Inf. Syst. (TOIS)* 22 (1), 5–53.
- Herm, L.-V., Steinbach, T., Wanner, J., Janiesch, C., 2022. A nascent design theory for explainable intelligent systems. *Electron. Mark.* 32 (4), 2185–2205.
- Hohman, F., Kahng, M., Pienta, R., Chau, D.H., 2018. Visual analytics in deep learning: An interrogative survey for the next frontiers. *IEEE Trans. Vis. Comput. Graph. (TVCG)* 25 (8), 2674–2693.
- Holter, S., El-Assady, M., 2024. Deconstructing human-AI collaboration: Agency, interaction, and adaptation. In: *Computer Graphics Forum*. Vol. 43, Wiley Online Library, e15107.
- Huertas, A., 2018. How live videos and stories in social media influence tourist opinions and behaviour. *Inf. Technol. Tour.* 19 (1), 1–28.
- Iren, D., Bilgen, S., 2014. Cost of quality in crowdsourcing. *Hum. Comput.* 1 (2).
- Jannach, D., Zanker, M., Felfernig, A., Friedrich, G., 2010. *Recommender Systems: An Introduction*. Cambridge University Press.
- Jiang, Y., Leiva, L.A., Rezazadegan Tavakoli, H., RB Houssel, P., Kylmäla, J., Oulasvirta, A., 2023. UEyes: Understanding visual saliency across user interface types. In: *ACM Conference on Human Factors in Computing Systems*. CHI, pp. 1–21.
- Jin, W., Fan, J., Gromala, D., Pasquier, P., Hamarneh, G., 2021. EUCA: The end-user-centered explainable AI framework. *arXiv preprint arXiv:2102.02437*.
- Kaplan, K., 2019. 5 UX workshops and when to use them: A cheat sheet. <https://www.nngroup.com/articles/5-ux-workshops/>.
- Knijnenburg, B.P., Willemsen, M.C., Gantner, Z., Soncu, H., Newell, C., 2012. Explaining the user experience of recommender systems. *User Model. User-Adapt. Interact.* 22, 441–504.
- Ko, H., Lee, S., Park, Y., Choi, A., 2022. A survey of recommendation systems: recommendation models, techniques, and application fields. *Electronics* 11 (1), 141.
- Kouki, P., Schaffer, J., Pujara, J., O'Donovan, J., Getoor, L., 2020. Generating and understanding personalized explanations in hybrid recommender systems. *ACM Trans. Interact. Intell. Syst. (TIIS)* 10 (4), 1–40.
- Ledwich, M., Zaitsev, A., 2019. Algorithmic extremism: Examining YouTube's rabbit hole of radicalization. *arXiv preprint arXiv:1912.11211*.
- Liao, Q.V., Gruen, D., Miller, S., 2020. Questioning the AI: informing design practices for explainable AI user experiences. In: *ACM Conference on Human Factors in Computing Systems*. CHI, pp. 1–15.
- Liao, Q.V., Varshney, K.R., 2021. Human-centered explainable ai (xai): From algorithms to user experiences. *arXiv preprint arXiv:2110.10790*.
- Lubos, S., Tran, T.N.T., Felfernig, A., Polat Erdeniz, S., Le, V.-M., 2024. LLM-generated explanations for recommender systems. In: *ACM Conference on User Modeling, Adaptation and Personalization*. UMAP, pp. 276–285.
- Ma, Q., Ren, X., Huang, C., 2024. Xrec: Large language models for explainable recommendation. *arXiv preprint arXiv:2406.02377*.
- Miller, T., 2019. Explanation in artificial intelligence: Insights from the social sciences. *Artificial Intelligence* 267, 1–38.
- Milton, A., Ajmani, L., DeVito, M.A., Chancellor, S., 2023. "I see me here": Mental health content, community, and algorithmic curation on TikTok. In: *ACM Conference on Human Factors in Computing Systems*. CHI, pp. 1–17.
- Molnar, C., 2020. *Interpretable Machine Learning*.
- Murphy-Hill, E., Murphy, G.C., 2013. Recommendation delivery: Getting the user interface just right. In: *Recommendation Systems in Software Engineering*. Springer, pp. 223–242.
- Nanou, T., Lekakos, G., Fouskas, K., 2010. The effects of recommendations' presentation on persuasion and satisfaction in a movie recommender system. *Multimedia Syst.* 16, 219–230.
- Nguyen, T., Canossa, A., Zhu, J., 2024. How human-centered explainable ai interface are designed and evaluated: A systematic survey. *arXiv preprint arXiv:2403.14496*.
- Ooge, J., Kato, S., Verbert, K., 2022. Explaining recommendations in e-learning: Effects on adolescents' trust. In: *ACM Conference on Intelligent User Interfaces*. IUI, pp. 93–105.
- Polatidis, N., Georgiadis, C.K., 2013. Mobile recommender systems: An overview of technologies and challenges. In: *International Conference on Informatics & Applications*. ICIA, IEEE, pp. 282–287.
- Pu, P., Chen, L., 2006. Trust building with explanation interfaces. In: *Proceedings of the 11th International Conference on Intelligent User Interfaces*. pp. 93–100.
- Ribeiro, M.T., Singh, S., Guestrin, C., 2016. "Why should i trust you?" Explaining the predictions of any classifier. In: *ACM International Conference on Knowledge Discovery and Data Mining*. SIGKDD, pp. 1135–1144.
- Ricci, F., 2010. Mobile recommender systems. *Inf. Technol. Tour.* 12 (3), 205–231.
- Ricci, F., Rokach, L., Shapira, B., 2021. Recommender systems: Techniques, applications, and challenges. *Recomm. Syst. Handb.* 1–35.
- Shi, G., Deng, X., Luo, L., Xia, L., Bao, L., Ye, B., Du, F., Pan, S., Li, Y., 2024. LLM-powered explanations: Unraveling recommendations through subgraph reasoning. *arXiv preprint arXiv:2406.15859*.
- Silva, Í., Marinho, L., Said, A., Willemsen, M.C., 2024. Leveraging ChatGPT for automated human-centered explanations in recommender systems. In: *ACM Conference on Intelligent User Interfaces*. IUI, pp. 597–608.
- Silveira, T., Zhang, M., Lin, X., Liu, Y., Ma, S., 2019. How good your recommender system is? A survey on evaluations in recommendation. *Int. J. Mach. Learn. Cybern.* 10, 813–831.
- Springer, A., Whittaker, S., 2019. Progressive disclosure: empirically motivated approaches to designing effective transparency. In: *ACM Conference on Intelligent User Interfaces*. IUI, pp. 107–120.
- Sun, R., Li, X., Akella, A., Konstan, J.A., 2024. Large language models as conversational movie recommenders: A user study. *arXiv preprint arXiv:2404.19093*.
- Suresh, H., Gomez, S.R., Nam, K.K., Satyanarayan, A., 2021. Beyond expertise and roles: A framework to characterize the stakeholders of interpretable machine learning and their needs. In: *ACM Conference on Human Factors in Computing Systems*. CHI, pp. 1–16.
- Szymanski, M., Ooge, J., De Croon, R., Vanden Abeele, V., Verbert, K., 2024. Feedback, control, or explanations? supporting teachers with steerable distractor-generating AI. In: *International Conference on Learning Analytics and Knowledge Conference*. LAK, pp. 690–700.
- Tintarev, N., Masthoff, J., 2007. A survey of explanations in recommender systems. In: *International Conference on Data Engineering*. ICDE, IEEE, pp. 801–810.
- Tintarev, N., Masthoff, J., 2012. Beyond explaining single item recommendations. In: *Recommender Systems Handbook*. Springer, pp. 711–756.
- Tintarev, N., Masthoff, J., 2015. Explaining recommendations: Design and evaluation. In: *Recommender Systems Handbook*. Springer, pp. 353–382.

- Van Someren, M., Barnard, Y.F., Sandberg, J., 1994. The think aloud method: a practical approach to modelling cognitive. Lond.: Acad. 11 (6).
- Vig, J., Sen, S., Riedl, J., 2009. Tagsplanations: explaining recommendations using tags. In: ACM Conference on User Modeling, Adaptation and Personalization. UMAP, pp. 47–56.
- Voramontri, D., Klieb, L., 2019. Impact of social media on consumer behaviour. Int. J. Inf. Decis. Sci. 11 (3), 209–233.
- Vultureanu-Albiși, A., Bădică, C., 2021. Recommender systems: An explainable AI perspective. In: International Conference on Innovations in Intelligent Systems and Applications. INISTA, IEEE, pp. 1–6.
- Wang, D., Yang, Q., Abdul, A., Lim, B.Y., 2019. Designing theory-driven user-centric explainable AI. In: ACM Conference on Human Factors in Computing Systems. CHI, pp. 1–15.
- Zhang, Y., Chen, X., et al., 2020. Explainable recommendation: A survey and new perspectives. Found. Trends® Inf. Retr. 14 (1), 1–101.